Junior Mathematical Challenge – 2010

1. What is 2010 + (+2010) + (−2010) − (+2010) − (−2010) ?

A 0 B 2010 C 4020 D 6030 E 8040

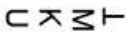
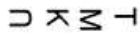
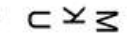
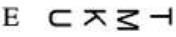
1041 ©UKMT

1. **B** The expression = 2010 + 2010 − 2010 − 2010 + 2010
= (2010 − 2010) + (2010 − 2010) + 2010 = 2010.

(https://ukmt.org.uk)

2. Each letter in the abbreviation shown is rotated through 90° clockwise. Which of the following could be the result?

U K M T

A  B  C D  E 

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2. **E** In A, the letter T is incorrect; in B it is U which is incorrect; in C and D the incorrect letters are M and K respectively.

3. Which of the following could have a length of 2010 mm?

A a table B an oil tanker C a teaspoon D a school hall E a hen's egg

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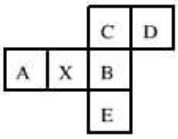
3. **A** 2010 mm = 2.01 m so, of the alternatives given, only a table could be expected to have a length of 2010 mm.

4. If the net shown is folded to make a cube, which letter is opposite X ?

A B C D E



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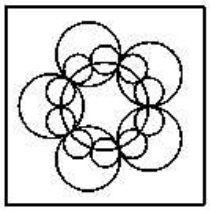
4. D Let X be on the top face of the cube. If the base is placed on a horizontal surface, then A, B, C, E will all be on vertical faces of the cube and D will be on the base, opposite X.

5. The diagram shows a pattern of 16 circles inside a square.
The central circle passes through the points where the other circles touch.
The circles divide the square into regions. How many regions are there?

A 17 B 26 C 30 D 32 E 38



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5. D Each of the five outer circles is divided into six regions, giving 30 regions in total. In addition, there is one region in the centre of the diagram and one region between the circles and the sides of the square. So, in all, there are 32 regions.

6. Which of the following has the largest value?

A $6 \div \frac{1}{2}$ B $5 \div \frac{1}{3}$ C $4 \div \frac{1}{4}$ D $3 \div \frac{1}{5}$ E $2 \div \frac{1}{6}$

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6. **C** The values of the expressions are A 12; B 15; C 16; D 15 and E 12.

7. Mr Owens wants to keep the students quiet during a Mathematics lesson. He asks them to multiply all the numbers from 1 to 99 together and then tell him the last-but-one digit of the result. What is the correct answer?

A 0 B 1 C 2 D 8 E 9

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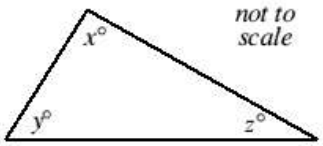
7. **A** As 2, 5 and 10 are all factors of the correct product, this product is a multiple of 100. So the last digit and the last-but-one digit are both zero.

8. In a triangle with angles $x°$, $y°$, $z°$ the mean of y and z is x.
What is the value of x?

A 90 B 80 C 70 D 60 E 50



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8. D If the mean of y and z is x, then $y + z = 2x$. So the sum of the interior angles of the triangle is $(x + y + z)° = 3x°$. So $3x = 180$, that is $x = 60$.

9. Which of the following is the longest period of time?

A 3002 hours B 125 days C $17\frac{1}{2}$ weeks D 4 months E $\frac{1}{3}$ of a year

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9. A One year is, at most, 366 days, so one-third of a year is less than 125 days. No month is longer than 31 days, so 4 months is also less than 125 days, as is 17.5 weeks which equals 122.5 days. However 3002 hours equals 125 days 2 hours, so this is the longest of the five periods of time.

10. At the Marldon Apple-Pie-Fayre bake-off, prize money is awarded for 1st, 2nd and 3rd places in the ratio 3 : 2 : 1. Last year Mrs Keat and Mr Jewell shared third prize equally. What fraction of the total prize money did Mrs Keat receive?

A $\frac{1}{4}$ B $\frac{1}{5}$ C $\frac{1}{6}$ D $\frac{1}{10}$ E $\frac{1}{12}$

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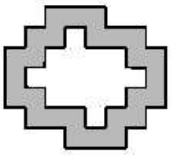
10. E Third prize is worth one-sixth of the total prize money, so Mrs Keat received half of that amount, that is one-twelfth of the total.

11. In the diagram shown, all the angles are right angles and all the sides are of length 1 unit, 2 units or 3 units.
What, in square units, is the area of the shaded region?



A 22 B 24 C 26 D 28 E 30

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11. C Divide the whole figure into horizontal strips of height 1 unit: its area is $(3 + 6 + 8 + 8 + 8 + 6 + 3)$ units^2 = 42 units^2. Similarly, the unshaded area is $(1 + 4 + 6 + 4 + 1)$ units^2 = 16 units^2. So the shaded area is 26 units^2.
Alternative solution: notice that if the inner polygon is moved a little, the answer remains the same – because it is just the difference between the areas of the two polygons. So, although we are not told it, we may assume that the inner one is so positioned that the outer shaded area can be split neatly into 1 by 1 squares – and there are 26 of these.

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12. Sir Lance has a lot of tables and chairs in his house. Each rectangular table seats eight people and each round table seats five people. What is the smallest number of tables he will need to use to seat 35 guests and himself, without any of the seating around these tables remaining unoccupied?

A 4 B 5 C 6 D 7 E 8

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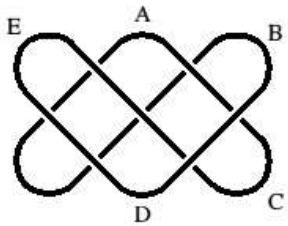
12. C There are 36 people to be seated so at least five tables will be required. The number of circular tables must be even. However, five rectangular tables will seat 40 people and three rectangular and two circular will seat 34. So at least six tables are needed. Two rectangular and four circular tables do seat 36 people: so six is the minimum number of tables.

13. The diagram shows a Lusona, a sand picture of the Tshokwe people from the West Central Bantu area of Africa. To draw a Lusona the artist uses a stick to draw a single line in the sand, starting and ending in the same place without lifting the stick in between. At which point could this Lusona have started?



A B C D E

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13. B It is necessary to find a route for which the line is broken the first time it passes through any intersection and solid when it passes through that intersection for the second time. Only the route which starts at B and heads away from D satisfies this condition.

14. The Severn Bridge has carried just over 300 million vehicles since it was opened in 1966. On average, roughly how many vehicles is this per day?

A 600 B 2 000 C 6 000 D 20 000 E 60 000

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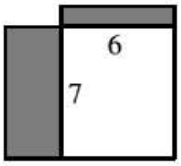
14. D The average number of vehicles per day $\approx \dfrac{300\,000\,000}{44 \times 365} \approx \dfrac{300\,000\,000}{40 \times 400}$

$= \dfrac{300\,000\,000}{16\,000} \approx \dfrac{300\,000\,000}{15\,000} = 20\,000.$

15. A 6 by 8 and a 7 by 9 rectangle overlap with one corner coinciding as shown.
What is the area (in square units) of the region *outside* the overlap?

A 6 B 21 C 27 D 42 E 69



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15. C The two shaded regions measure 3 by 7 and 1 by 6, so the total area outside the overlap is 27 units^2.

16. One of the examination papers for Amy's Advanced Arithmetic Award was worth 18% of the final total. The maximum possible mark on this paper was 108 marks. How many marks were available overall?

A 420 B 480 C 540 D 560 E 600

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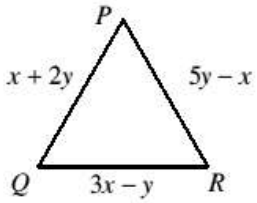
16. E As 108 marks represented 18% of the final total, 6 marks represented 1% of the final total. So this total was 600.

17. The lengths, in cm, of the sides of the equilateral triangle PQR are as shown.
Which of the following could *not* be the values of x and y?

A (18, 12) B (15, 10) C (12, 8) D (10, 6) E (3, 2)



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17. D As triangle PQR is equilateral, $x + 2y = 3x - y = 5y - x$. Equating any two of these expressions gives $2x = 3y$.
The only pair of given values which does not satisfy this equation is $x = 10$, $y = 6$.

(https://ukmt.org.uk)

18. Sam's 101st birthday is tomorrow. So Sam's age in years changes from a square number (100) to a prime number (101). How many times has this happened before in Sam's lifetime?

A 1 B 2 C 3 D 4 E 5

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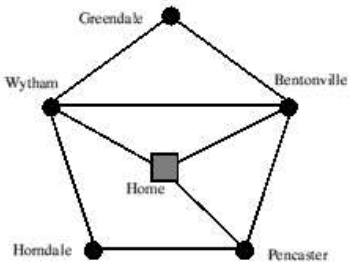
18. D The other times that this has happened previously are when Sam's age in years went from 1 to 2; from 4 to 5; from 16 to 17 and from 36 to 37.
Note that since primes other than 2 are odd, the only squares which need to be checked, other than 1, are of even numbers.

19. Pat needs to travel down every one of the roads shown at least once, starting and finishing at home. What is the smallest number of the five villages that Pat will have to visit more than once?

A 1 B 2 C 3 D 4 E 5



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19. C Villages which have more than two roads leading to them (or from them) must all be visited more than once as a single visit will involve at most two roads. So Bentonville, Pencaster and Wytham must all be visited more than once. The route Home, Bentonville, Greendale, Wytham, Bentonville, Pencaster, Home, Wytham, Horndale, Pencaster, Home starts and finishes at Home and visits both Greendale and Horndale exactly once so the minimum number of villages is three.

20. Nicky has to choose 7 different positive whole numbers whose mean is 7. What is the largest possible such number she could choose?

A 7 B 28 C 34 D 43 E 49

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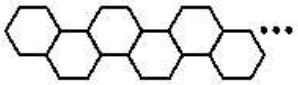
20. B The seven numbers must total 49 if their mean is to be 7. The largest possible number will occur when the other six numbers are as small as possible, that is 1, 2, 3, 4, 5, 6. So the required number is $49 - 21 = 28$.

21. A shape consisting of a number of regular hexagons is made by continuing to the right the pattern shown in the diagram, with each extra hexagon sharing one side with the preceding one. Each hexagon has a side length of 1 cm. How many hexagons are required for the perimeter of the whole shape to have length 2010 cm?



A 335 B 402 C 502 D 670 E 1005

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21. C The first and last hexagons both contribute 5 cm to the perimeter of the pattern. Every other hexagon in the pattern contributes 4 cm to the perimeter. The first and last thus contribute 10 cm, so we need another $2000 \div 4 = 500$ hexagons. Therefore the total number of hexagons required is 502.

22. Kiran writes down six different prime numbers, p, q, r, s, t, u, all less than 20, such that $p + q = r + s = t + u$. What is the value of $p + q$?

A 16 B 18 C 20 D 22 E 24

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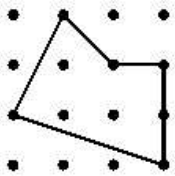
22. E The prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19. It is not possible for 2 to be one of the six numbers Kiran wrote down, since that would give one of the pairs an odd sum, whereas both of the other pairs would add up to an even number. The sum of the remaining 7 primes is 75 which is a multiple of 3. The sum of the six primes making up the three pairs must also be a multiple of 3 since each pair has the same total. So the odd prime not used in the six pairs must be a multiple of 3 too. Therefore 3 is the odd prime not used. So each pair totals $72 \div 3$, that is 24, and the pairs are $5 + 19$, $7 + 17$, $11 + 13$.

23. A single polygon is made by joining dots in the 4×4 grid with straight lines, which meet only at dots at their end points. No dot is at more than one corner. The diagram shows a five-sided polygon formed in this way. What is the greatest possible number of sides of a polygon formed by joining the dots using these same rules?



A 12 B 13 C 14 D 15 E 16

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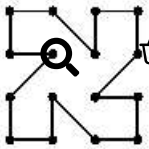
(https://ukmt.org.uk)

23. E The number of sides of the polygon is equal to the number of corners it has. As no dot is at more than one corner, the maximum number of corners is 16. So the maximum possible number of sides is 16, provided that a 16-sided figure may be drawn. The figure on the right shows one of several ways in which this can be achieved.



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24. The year 2010 belongs to a special sequence of twenty-five consecutive years: each number from 1988 to 2012 contains a repeated digit.

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Half of the following belongs to a sequence of consecutive years, where each number in the sequence contains at least one repeated digit.

Which of them belongs to the next such sequence of at least twenty years?

A 2099 B 2120 C 2199 D 2989 E 3299

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25. What is the value of $P + Q + R$ in the multiplication on the right?

A 13 B 12 C 11 D 10 E 9

$$\begin{array}{cccccc} & & P & Q & P & Q \\ \times & & & R & R & R \\ \hline 6 & 3 & 9 & 0 & 2 & 7 \end{array}$$

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The three-digit number RRR is equal to 111 multiplied by the single digit R. So $PQPQ \times R = 639027 \div 111 = 5757$. Now $PQPQ$ equals the two-digit number PQ multiplied by 101. So $PQ \times R = 5757 \div 101 = 57$. The only ways in which 57 may be expressed as the product of a two-digit number and a single digit are 57×1 and 19×3. So $P = 5, Q = 7, R = 1$ or $P = 1, Q = 9, R = 3$. In both cases, $P + Q + R = 13$.

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